AWS VPC Peering Project

Build, validate, and explain private connectivity between two AWS VPCs

Student	Region	Completion date
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Project outcome

This report documents a two-VPC AWS lab with a public bastion host, two private EC2 workloads, VPC peering, explicit route tables, scoped security groups, and verified private connectivity.

VPC A and VPC B were connected through an active peering connection. App-A and App-B remained private, while administration was performed through a single bastion host in VPC A. Bidirectional ICMP testing succeeded with zero packet loss after routes and security rules were configured.

1 Architecture and resource plan

The design uses non-overlapping IPv4 CIDRs. VPC A contains a public subnet for the bastion and a private subnet for App-A. VPC B contains a private subnet for App-B and has no internet gateway.

Component	Name	Address or purpose
VPC A	ewb-vpc-a	10.10.0.0/16
Public subnet	ewb-a-public	10.10.1.0/24 | Bastion 10.10.1.10
Private subnet	ewb-a-private	10.10.2.0/24 | App-A 10.10.2.10
VPC B	ewb-vpc-b	10.20.0.0/16
Private subnet	ewb-b-private	10.20.1.0/24 | App-B 10.20.1.10
Peering	ewb-a-b-peer	Private connectivity between VPC A and VPC B

Traffic flow

Laptop -> Bastion 10.10.1.10 -> App-A 10.10.2.10 <-> VPC Peering <-> App-B 10.20.1.10

The laptop reaches only the bastion over SSH. ProxyJump makes the bastion create the private SSH connection to App-A or App-B. Application traffic uses private IPv4 addresses over the peering connection.

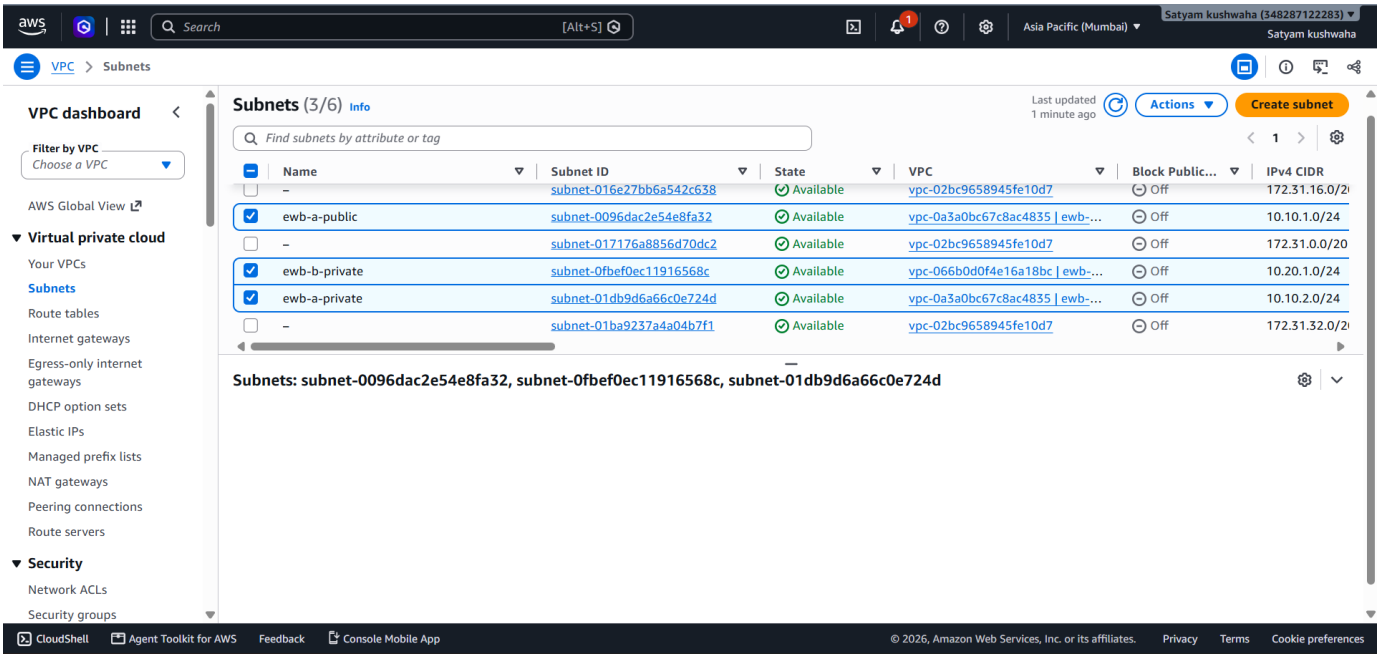


Figure 1. The final subnet inventory shows the three project subnets and their non-overlapping CIDRs.

2 Resource inventory

Resource	Name	Actual identifier
VPC	ewb-vpc-a	vpc-0a3a0bc67c8ac4835
VPC	ewb-vpc-b	vpc-066b0d0f4e16a18bc
Peering connection	ewb-a-b-peer	pcx-060ec3cd3fa6cb947
Route table	ewb-rt-a-public	rtb-0d30fe1780f32d940
Route table	ewb-rt-a-private	rtb-06f68f9c778b8f218
Route table	ewb-rt-b-private	rtb-0aa75c8f70d9bf283
EC2	ewb-bastion	i-00158c098d37d1d0e
EC2	ewb-app-a	i-05bfba147a5f3bcf6
EC2	ewb-app-b	i-0f1d4742abc2fb3e4

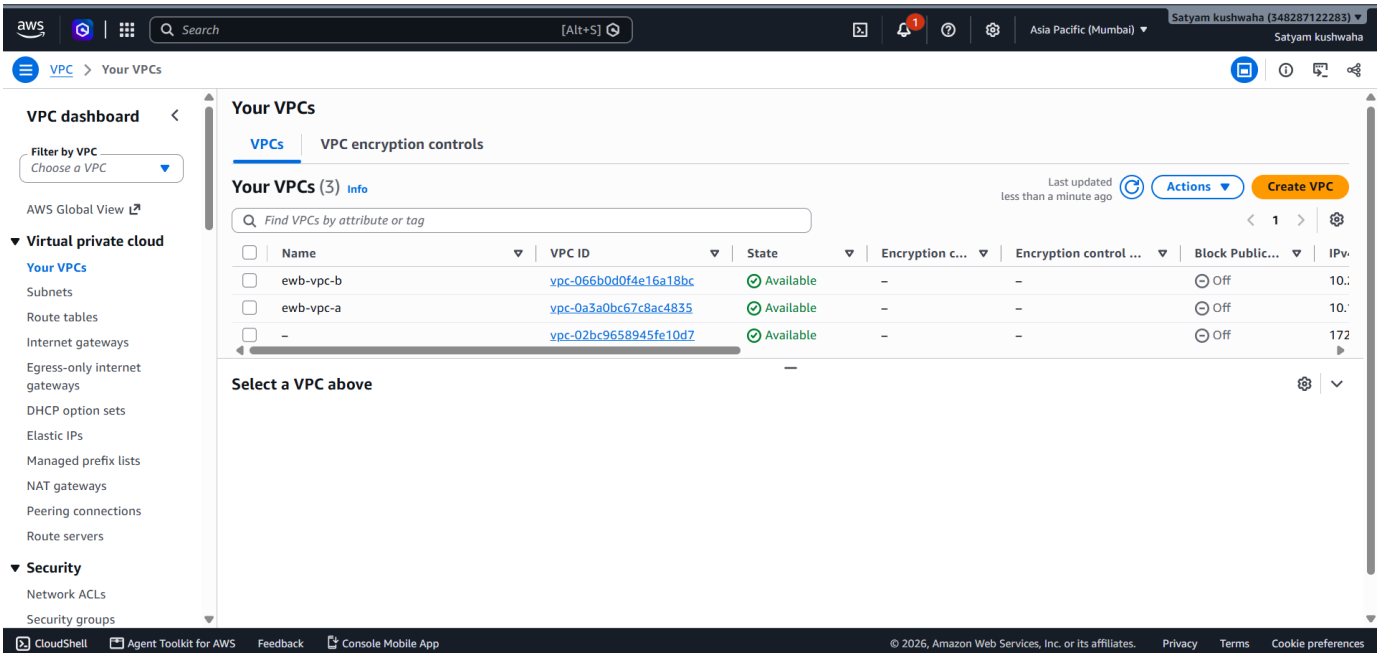


Figure 2. Both project VPCs are available in the Mumbai Region.

3 Route tables and subnet associations

Each subnet is explicitly associated with a project route table. VPC A public routing includes the internet gateway and a peering route; the two private route tables include only their local route plus the required peer route.

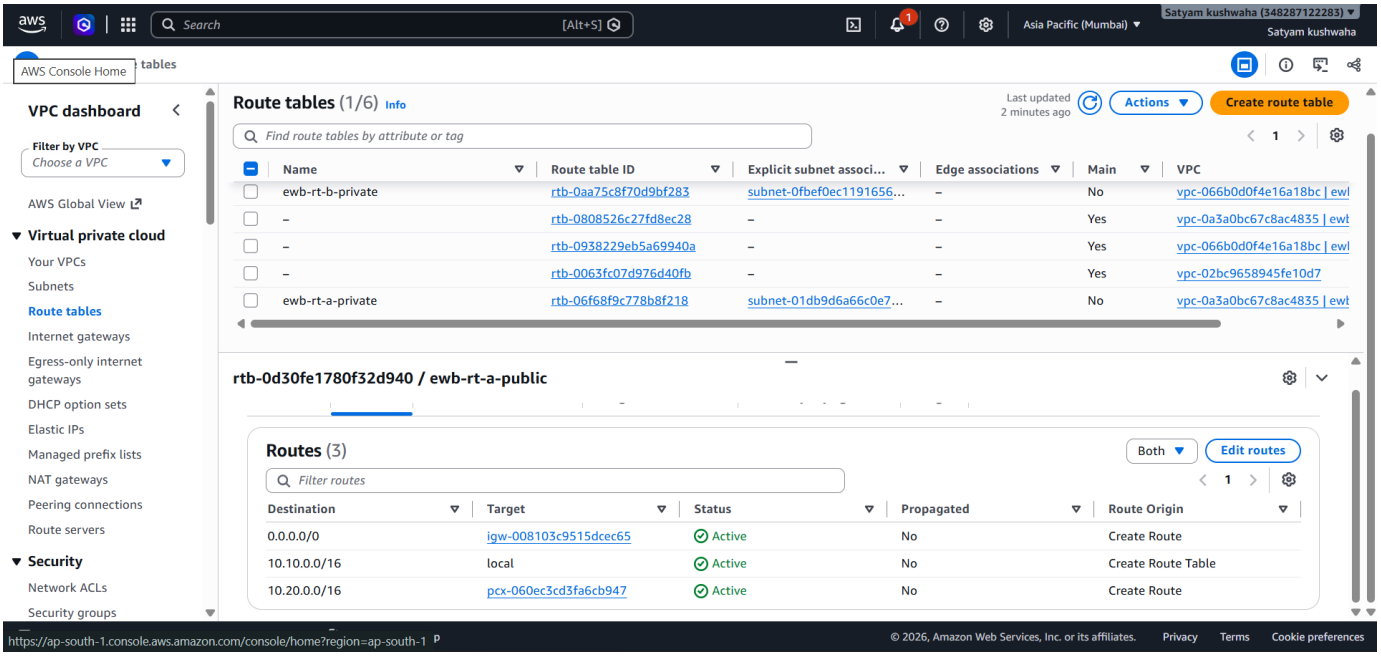


Figure 3. ewb-rt-a-public contains the internet gateway default route and the route to VPC B through peering.

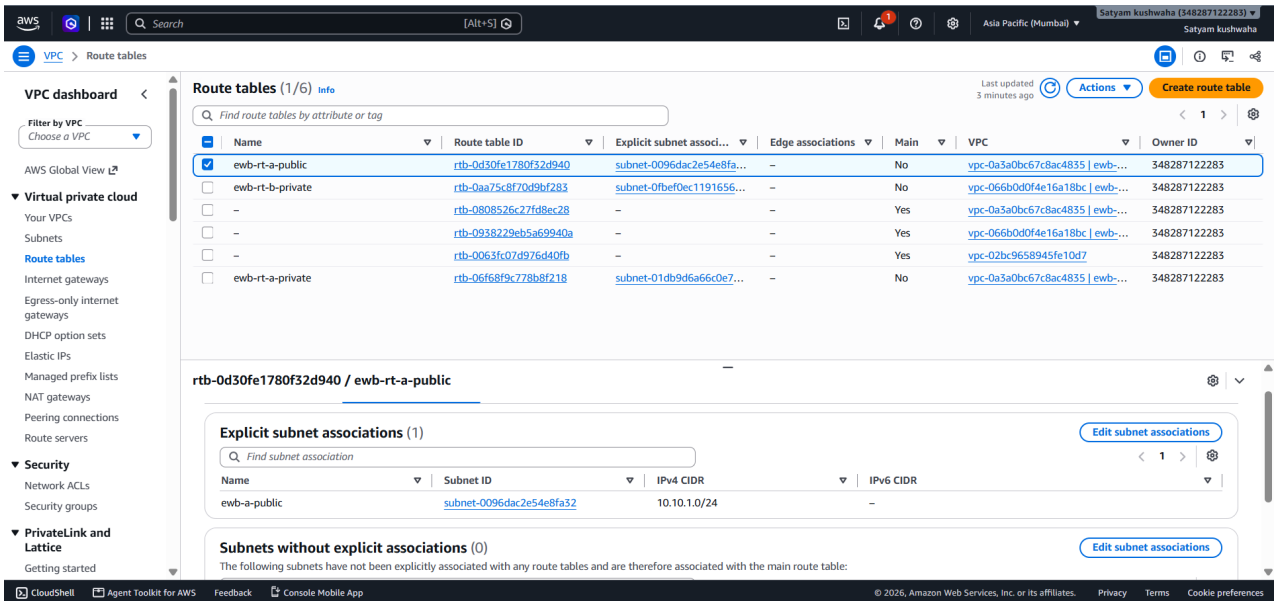


Figure 4. ewb-a-public is explicitly associated with ewb-rt-a-public.

3 Route tables and subnet associations continued

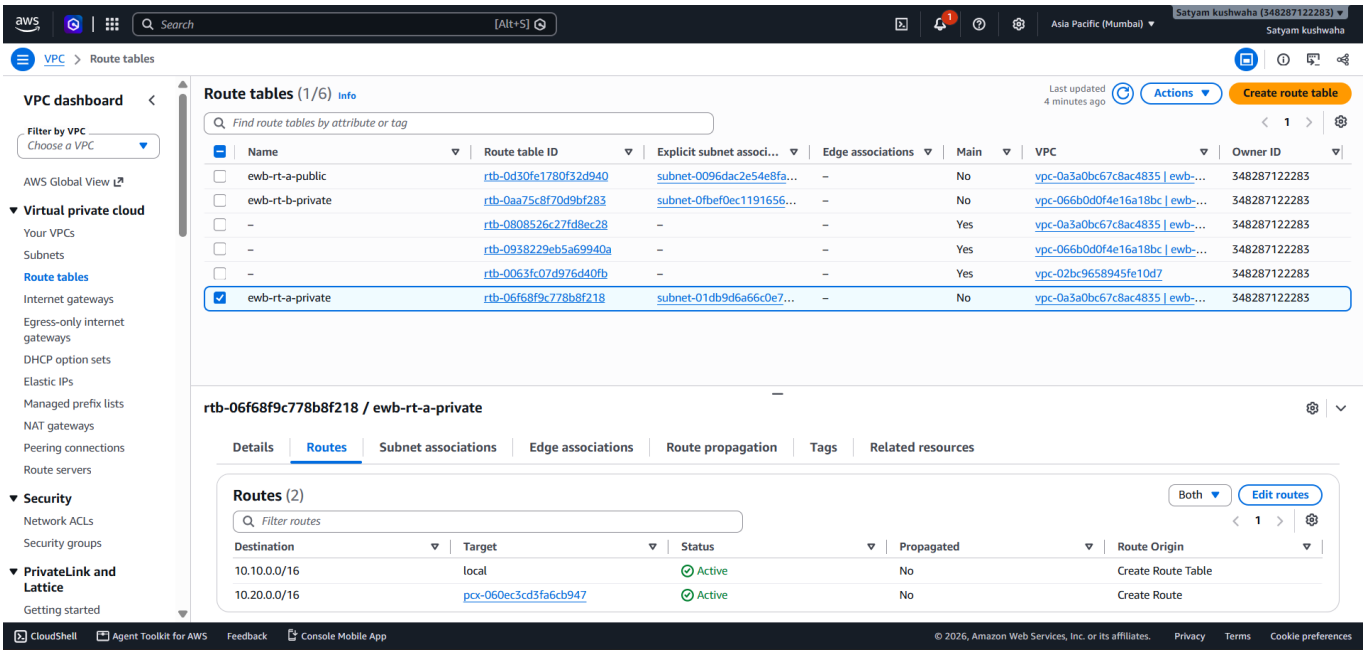


Figure 5. ewb-rt-a-private routes traffic for 10.20.0.0/16 to the peering connection.

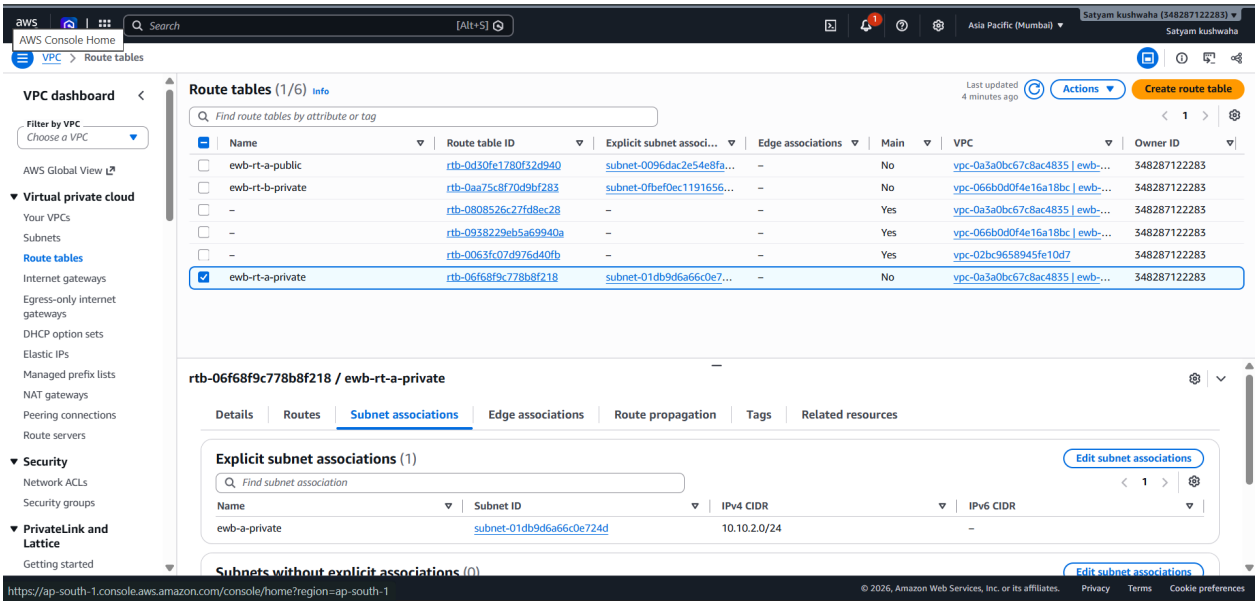


Figure 6. ewb-a-private is explicitly associated with ewb-rt-a-private.

3 Route tables and subnet associations continued

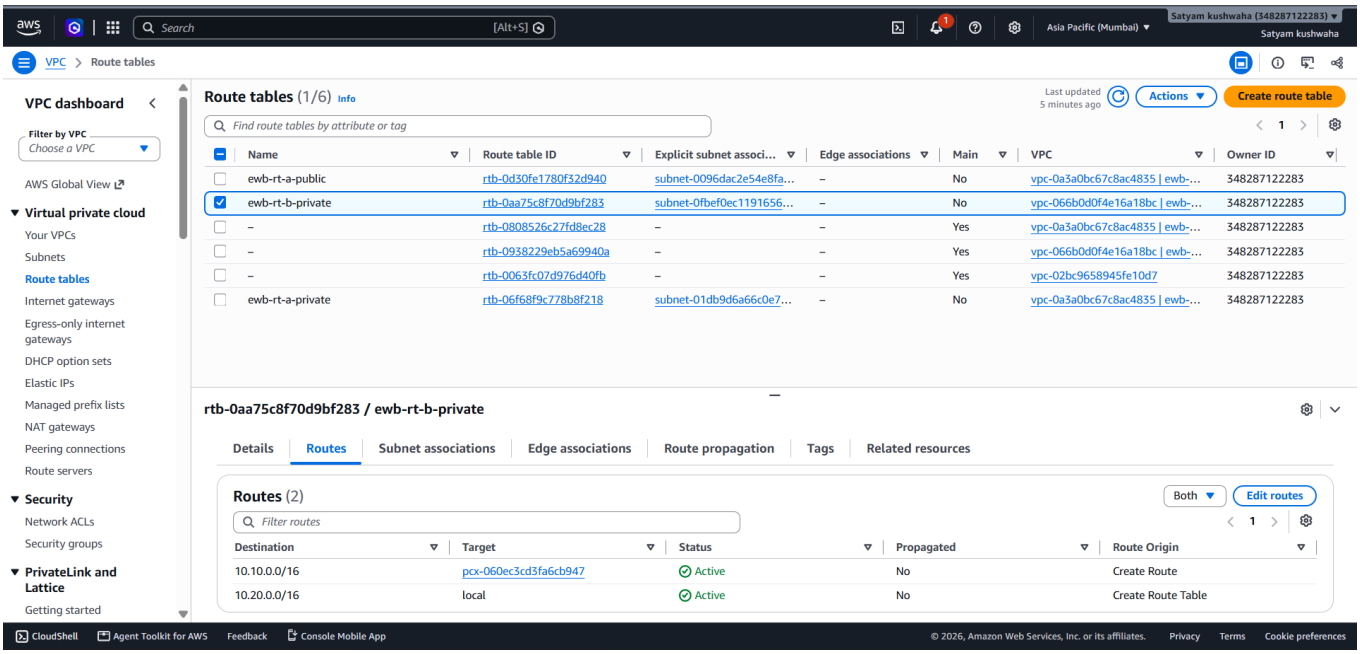


Figure 7. ewb-rt-b-private provides the return route for 10.10.0.0/16 through peering.

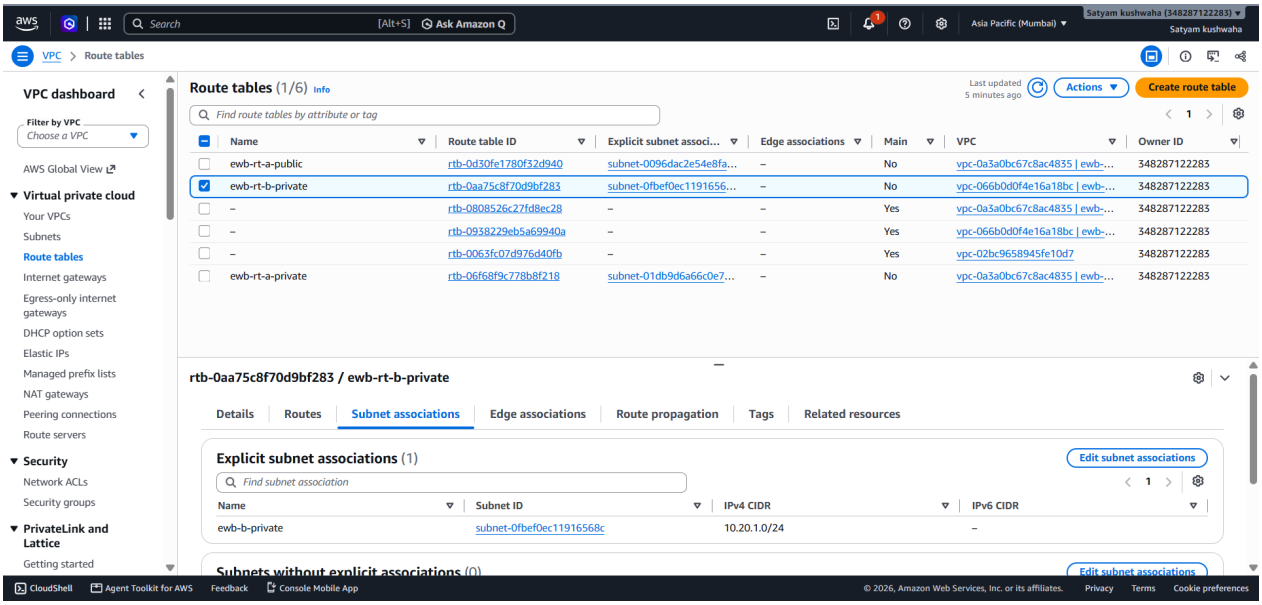


Figure 8. ewb-b-private is explicitly associated with ewb-rt-b-private.

4 VPC peering

The peering connection is active between VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16). The active connection alone is not sufficient for reachability; the route tables and security groups provide the required path and permissions.

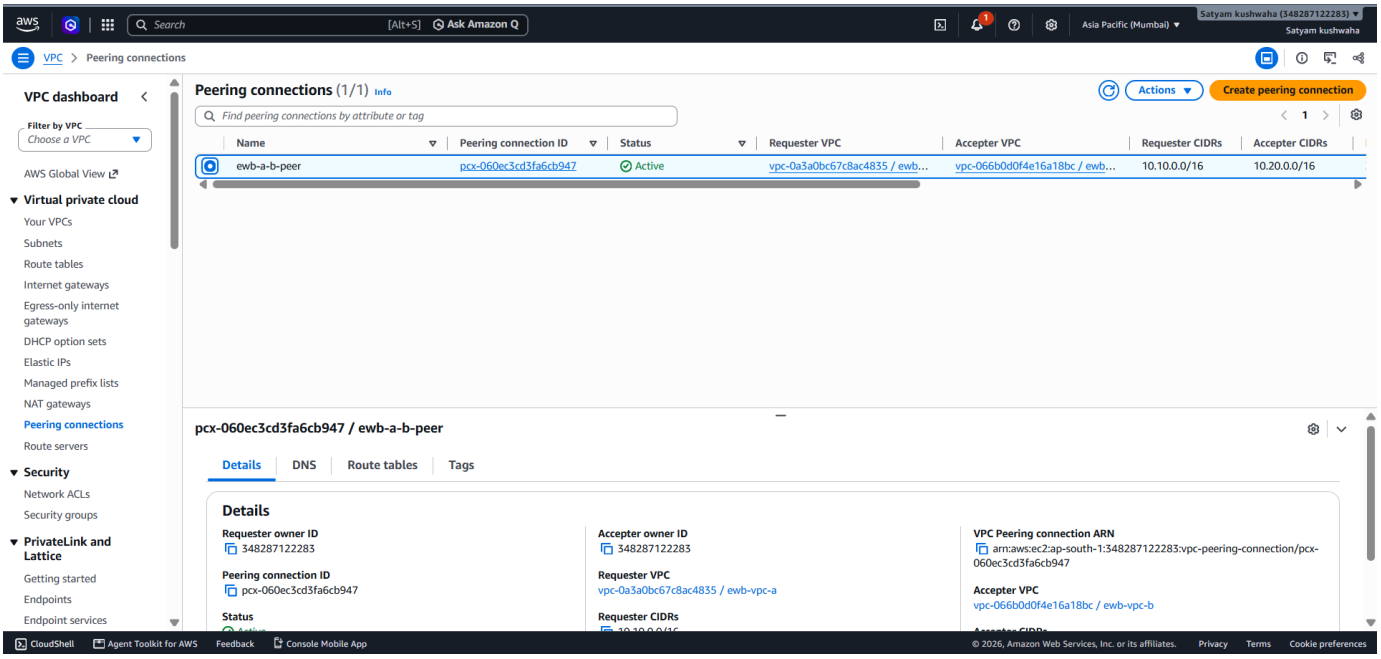


Figure 9. ewb-a-b-peer is active between the requester and accepter VPCs.

5 Security controls

Inbound access is limited to the required source addresses. The bastion accepts SSH only from the student public IP. App-A and App-B accept SSH only from the bastion private IP and ICMP echo requests only from the peer application host.

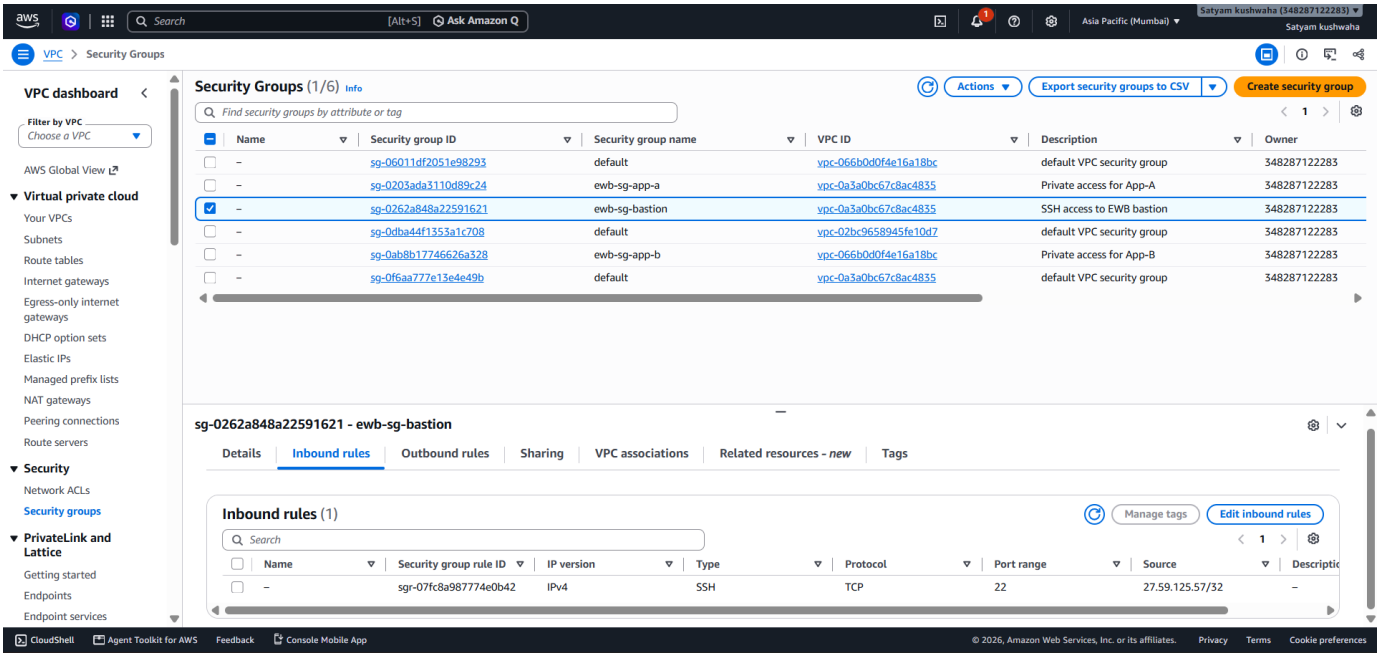


Figure 10. Bastion security group permits SSH on TCP 22 from one public /32 address.

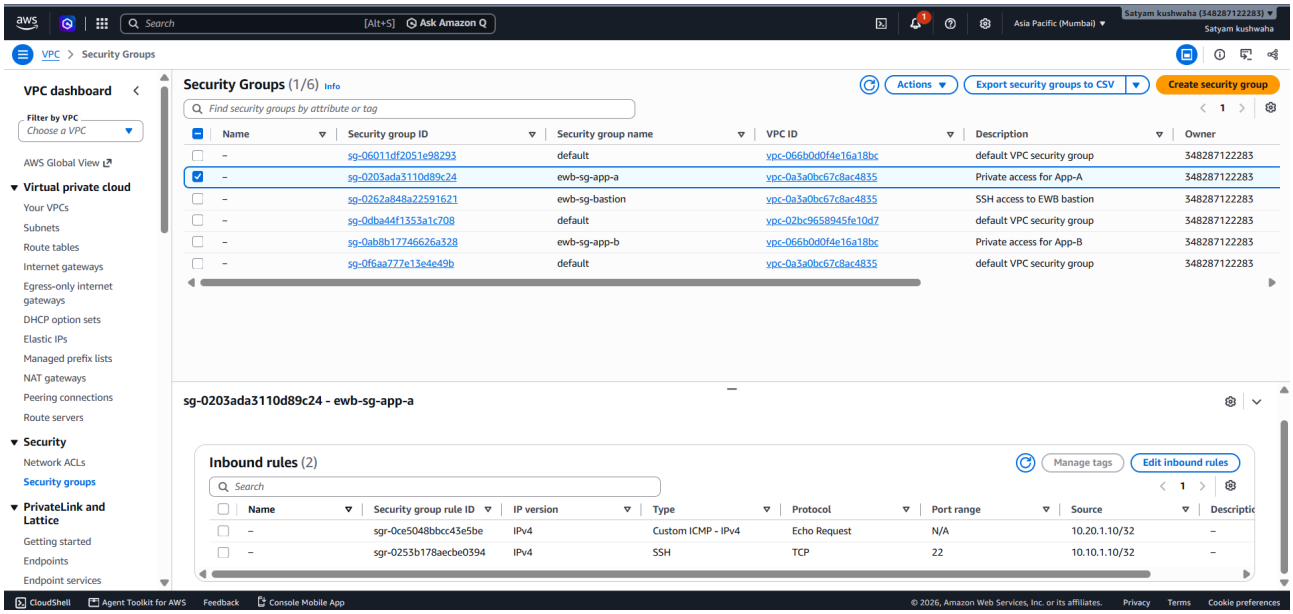


Figure 11. App-A permits SSH from the bastion and ICMP echo requests from App-B.

5 Security controls continued

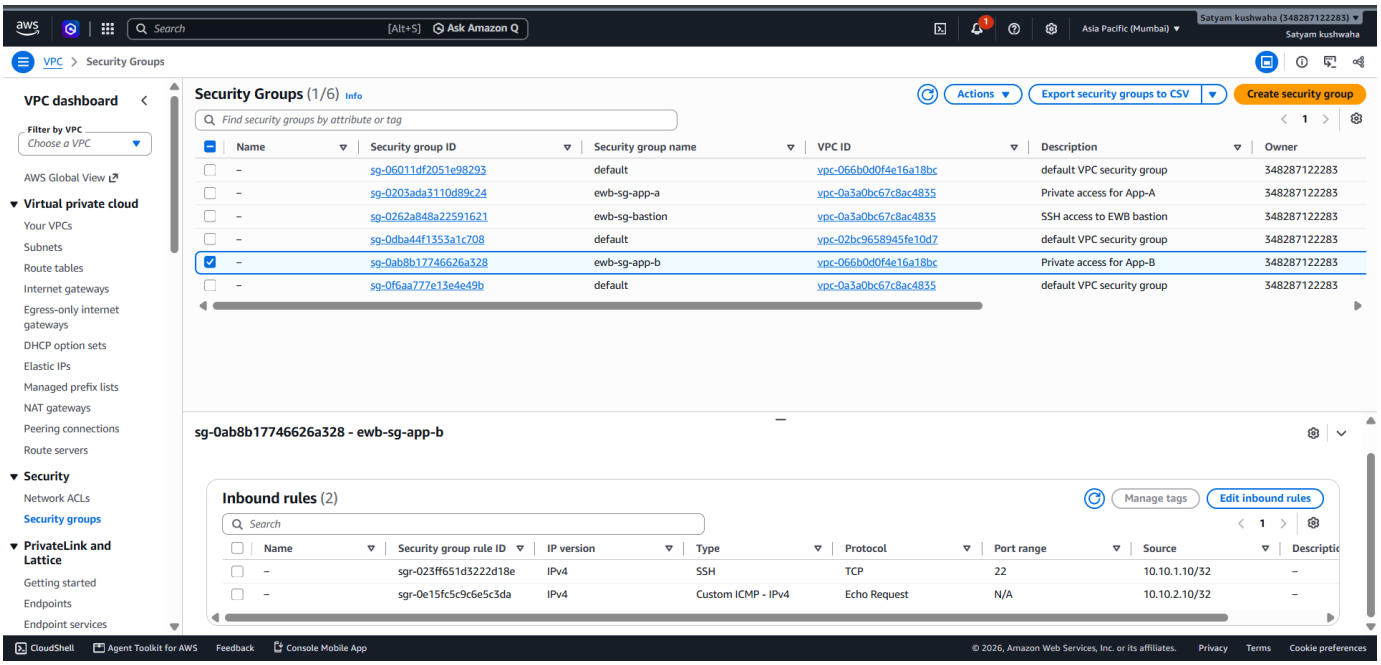


Figure 12. App-B permits SSH from the bastion and ICMP echo requests from App-A.

6 EC2 instance isolation

All three Amazon Linux 2023 instances are running and have passed status checks. The bastion has a public IPv4 address, while App-A and App-B show no public IPv4 address.

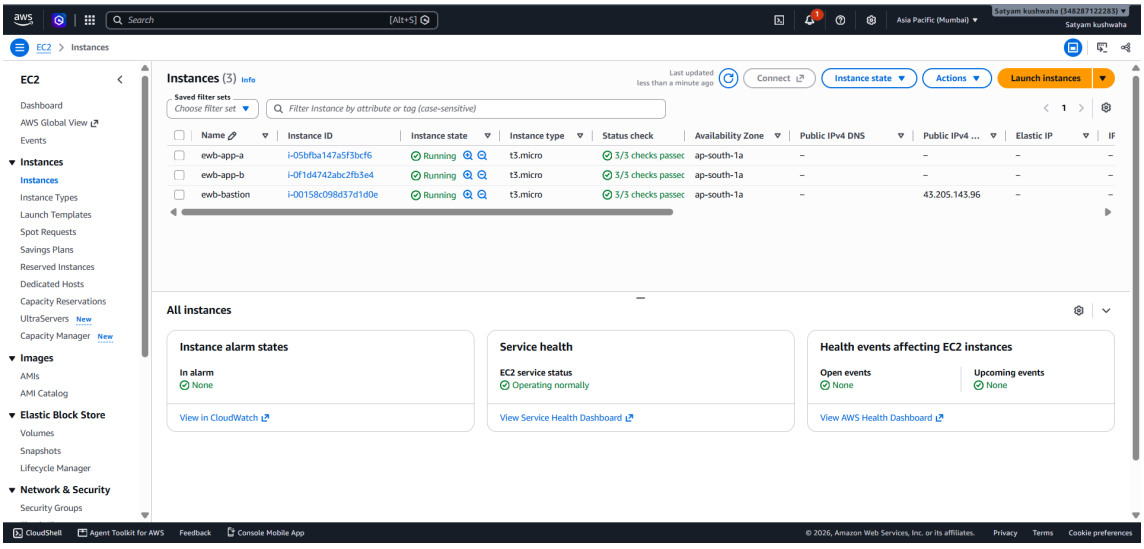


Figure 13. Instance inventory confirms private application hosts and one public bastion.

7 Connectivity validation

Administration was verified with SSH to the bastion and SSH ProxyJump to the private workloads. ICMP was then tested in both directions using private IPv4 addresses.

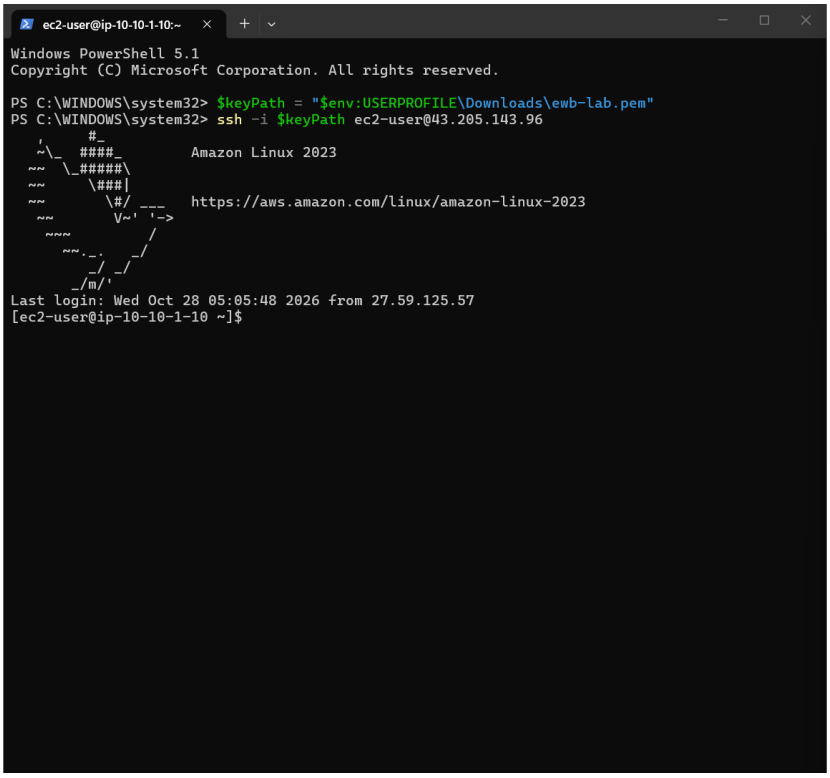


Figure 14. Successful SSH login to the public bastion at private address 10.10.1.10.

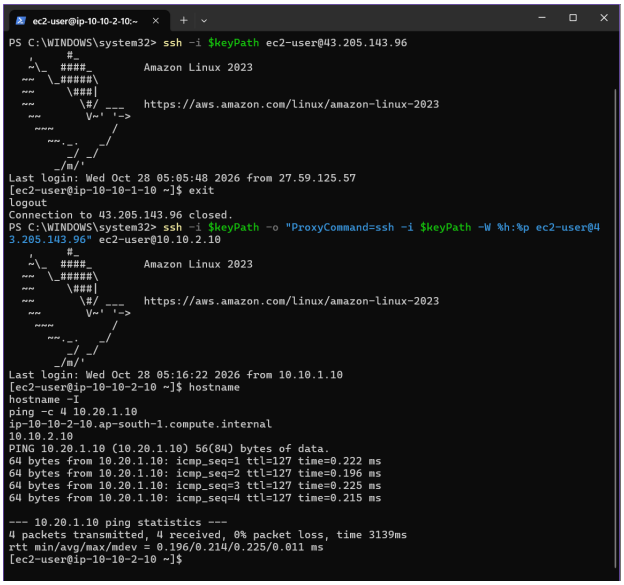


Figure 15. App-A (10.10.2.10) reaches App-B (10.20.1.10): 4 received, 0 percent packet loss.

7 Connectivity validation continued

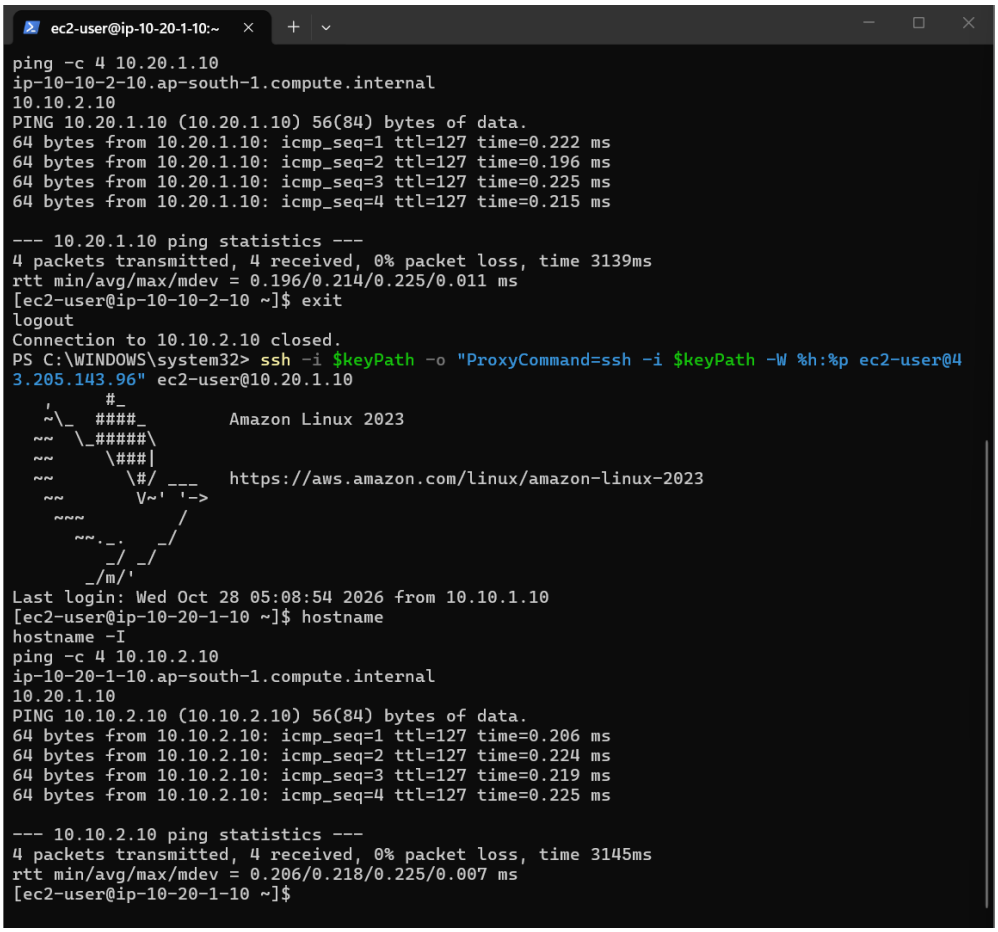


Figure 16. App-B (10.20.1.10) reaches App-A (10.10.2.10): 4 received, 0 percent packet loss.

8 Controlled failure and recovery

The App-B inbound ICMP echo rule for 10.10.2.10/32 was temporarily removed. The App-A ping then failed with 100 percent packet loss. After restoring the exact rule, the same ping succeeded with 4 received packets and zero packet loss.

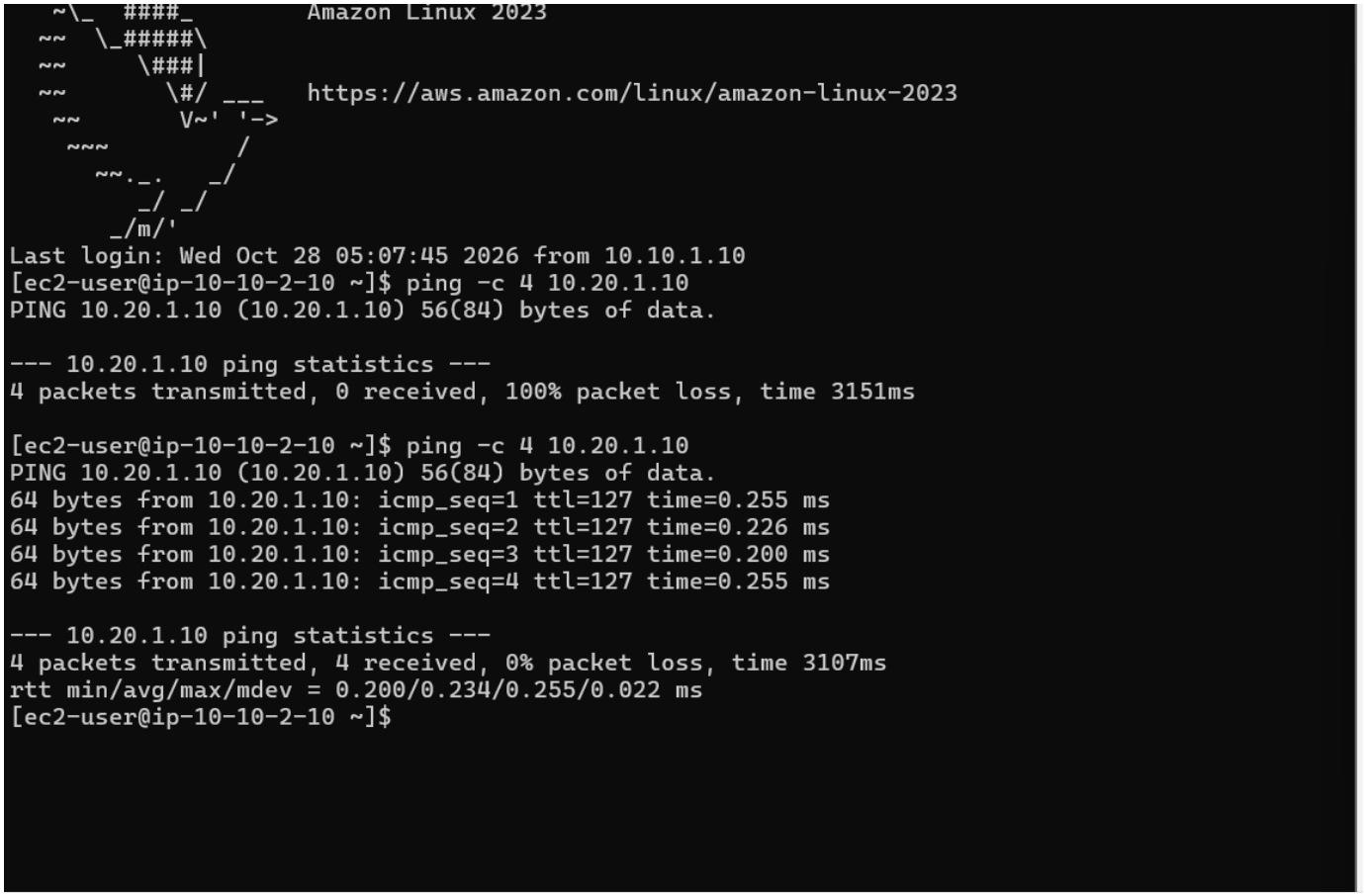


Figure 17. Controlled ICMP failure and recovery from App-A to App-B.

9 Explanation of the design

Why is the return route needed

A packet from App-A reaches App-B only if the VPC A private route table sends 10.20.0.0/16 through the peering connection. The response needs a matching route from VPC B back to 10.10.0.0/16. Without the return route, the reply cannot reach App-A.

Why does App-B see the bastion private IP for SSH

ProxyJump does not forward the laptop TCP connection directly to App-B. The bastion opens a separate SSH connection to App-B, so App-B sees the bastion private address 10.10.1.10 as the source.

Why does active peering not guarantee successful ping

Peering provides the connection, but not the route entries or security permissions. Both VPCs need correct routes, and the destination security group must permit ICMP echo requests from the source host.

Can VPC B use VPC A internet access through peering

No. VPC peering is non-transitive and does not provide shared internet gateway access. VPC B remains private and has no internet gateway in this design.

10 Cleanup

After assessment and report submission, terminate the three instances, remove peer routes, delete the peering connection, delete the security groups, delete custom route tables and subnets, detach and delete the internet gateway, and finally delete both lab VPCs. This prevents further charges.